

# Ali Tahir

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## PROFESSIONAL SUMMARY

SWE with 5+ years building production software and tooling across enterprise environments, plus five years running my own business. Focusing on applied AI and agentic systems: building MCP servers, agent workflows, persistent knowledge systems, and LLM-powered developer tools that move beyond prototypes into real team adoption. As comfortable owning critical infrastructure solo as I am leading a team's day-to-day delivery.

## SKILLS

### Agent-Focused Development

MCP servers, agent architecture, persistent agent memory, multi-agent workflows, model routing, structured knowledge systems, local and open-weight models, AI-assisted development

### Languages & Frameworks

TypeScript/JavaScript, Node.js, Python, C#/.NET, React, Django, FastAPI, Playwright, Chrome Extensions (Manifest V3)

### Spoken Languages

English (Native), Urdu/Hindi (Native), Punjabi (Receptive Fluency), Spanish (Conversational)

### Cloud & DevOps

GitHub Actions, AWS (ECS, EC2, CloudFormation, RDS, S3, IAM), Docker, Terraform, TeamCity

### Databases & APIs

SQL Server, PostgreSQL, SQLite, Metabase, GraphQL, REST APIs, Apigee

### Project Management

SAFe Agile with Jira, Kanban with Rally, Scrum, standup facilitation across two teams, sprint planning and retrospectives, end-to-end change management, cross-team reviews, stakeholder progress reporting

## PROFESSIONAL EXPERIENCE

### Industry Data Exchange Association (IDEA)

2025 – Present

#### Systems Developer II, Data Whispers Team

Multi-stack team building a customer-facing BI platform. Currently the **sole developer on core features**; established the delivery, deployment, and documentation processes.

- Removed a DBA bottleneck that cost hours per question by co-building **metabase-mcp**, a custom **MCP server** in TypeScript exposing **8 tools** that let AI agents query and update the company's analytics stack. Adopted **beyond engineering** by a second product team, the PM, and sales.
- Used the **MCP server** to cut troubleshooting on a complex, undocumented SQL Server database from **hours (sometimes days) to minutes** by generating a **queryable JSONL knowledge base**. Indexed **544 stored procedures** and **247 tables**, and mapped **3,670+ dependency relationships**, with every answer ranked to a confidence level using evidence labels checked against the live database.
- Leading a **ground-up rebuild of the .NET customer data ingestion service** that turns raw client files into the analytics records the platform sells. Spent a month extracting the undocumented process from the one teammate who carried it, then designed a **three-phase pipeline**: client-validated intake, rule-based sanitation to schema, automated ingestion.
- **Modernized the team's delivery infrastructure end to end**: migrated the dev team off AWS CodeCommit (**560+ branches**, no strategy) to GitHub with standardized PR workflows and training for the full team, and delivered the company's **first CI/CD pipelines** for two applications on distinct stacks, replacing manual releases with test-gated GitHub Actions deployment across **8 automated stages**.
- Built a repeatable capture practice that turns meetings, emails, and chats into **structured events** using a schema I designed. Results feed architecture guides, SOPs, runbooks, and a **queryable workflow history** across two product systems.

## General Electric Company / Accenture (embedded at GE, 2024–2025)

2021 – 2025

### Software Engineering Specialist / Security Specialist, SSO Team

Software engineer on the team owning end-to-end SDLC for 14 mission-critical SSO APIs across GE's historic split: GE Aerospace, GE Vernova, and GE Healthcare.

- Part of the team that architected and delivered **14 SSO APIs** in **11 months**, modernizing 15+ year legacy Java systems during simultaneous identity provider and stack migrations; **99% uptime** since production deployment, serving **80,000+ users** across three business units through GE's historic restructuring.
- **Independently operated the entire enterprise SSO infrastructure for 10 months as sole remaining developer** during the GE/Accenture transition, maintaining critical authentication services without interruption while running daily standups, sprint reviews, and retrospectives.
- Modernized security and delivery infrastructure: implemented mutual authentication (mTLS) with OpenSSL, Terraform, and Envoy to secure internal API traffic, and migrated CI/CD from Jenkins to TeamCity with custom build scripts and templates for **all three GE business units**.

## MAXX Potential

2021 – 2022

### Technology Apprentice

- Placed on a paid client engagement within **2 months** of hire and converted to a full-time GE software engineering role in **7 months**, against a typical 1–3 year apprenticeship track; also ran a 2-week program of instruction and Q&A onboarding a cohort of **12+ prospective apprentices** through MAXX Career Labs.

## HerbNCurry Catering Solutions

2015 – 2020

### Owner / Operator

Built and operated a specialized catering business from the ground up while completing an undergraduate degree.

- Grew from startup to **155 events annually** at peak, reaching **\$64,800 revenue** in the highest-earning year (2019) through client acquisition and operational discipline

## SELECTED PROJECTS

### AI Agent Harness Extensions

2026 – Present

- Designed **The Advisor**, a model-agnostic MCP server that implements **model re-routing as a spend and quality policy**: cheaper models carry the everyday workload, and only architectural or hard-to-reverse decisions escalate to a frontier model for an independent second opinion. Routing policy lives in server config, invisible to the calling agent, so it changes without touching agent code.
- Designed a **persistent project-memory system** that lets coding agents resume multi-session work without re-deriving context, with explicit rules for what gets recorded, where, and when. Validated through fresh-session testing against smaller models.

## EDUCATION & CERTIFICATIONS

AWS Cloud Practitioner (2023)

SAFe® Scrum Master (2023)

AWS Machine Learning Engineer (In Progress)

### Full-Stack Development Bootcamp

Coding Dojo, 2021

### B.S. - Experimental Psychology

University of Texas at San Antonio, 2018

Sponsored into graduate-level (5000) research: "Cross Cultural Stressors of Adolescents and Young Adults Ages 14-24 in a Bi-Cultural Upbringing"

**Community:** Meadowbridge Community Market (MAD RVA) volunteer since 2025; TechCrunch All Stage, Boston 2025